

DP Computer Science: B 1.1.4 Flowcharts

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Identify and explain the purpose of **standard flowchart symbols**:
 - Connector
 - Decision
 - Flowline
 - Input/Output
 - Process/Operation
 - Start/End
 - **Draw** and **interpret** flowcharts for a variety of programming algorithms
 - **Trace** the execution flow of a given flowchart, accurately following the sequence of steps
 - **Construct** and utilise a **trace table** to track changes in variable values during flowchart execution
 - **Determine** the output of an algorithm by tracing its flowchart
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Key Vocabulary

Term	Definition
Flowchart	A visual diagram that maps out the steps in an algorithm
Trace table	A table used to track changes in variable values as a flowchart is executed step by step
Algorithm	A step-by-step procedure for solving a problem
Execution flow	The sequence in which steps of an algorithm are performed
Iteration	The repetition of a set of instructions (looping)
Selection	A decision point in an algorithm (if-else)
Sequence	Steps performed in order, one after another

ATL Skills

Skill	Description
Thinking (Critical Thinking)	Analysing the logic of an algorithm represented in a flowchart
Information Literacy	Interpreting and extracting information from visual representations (flowcharts)
Problem Solving	Identifying and correcting errors in algorithm logic through flowchart tracing

Skill	Description
Communication	Clearly describing the execution flow of an algorithm using flowchart terminology
Self-Management	Systematically tracing the steps of a flowchart and recording variable changes

2. Introduction to Flowcharts

The Recipe Analogy

| "Imagine you're following a recipe. What are the key elements that guide you through the process?" (Ingredients, steps, decision points, etc.)

What is a Flowchart?

| A flowchart is a visual diagram that maps out the steps in an algorithm. It is like a recipe for a computer program.

Benefits of Flowcharts

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BENEFITS OF FLOWCHARTS

VISUALISING LOGIC

Makes complex logic easy to understand

IDENTIFYING ERRORS

Helps spot potential errors or inefficiencies

COMMUNICATING

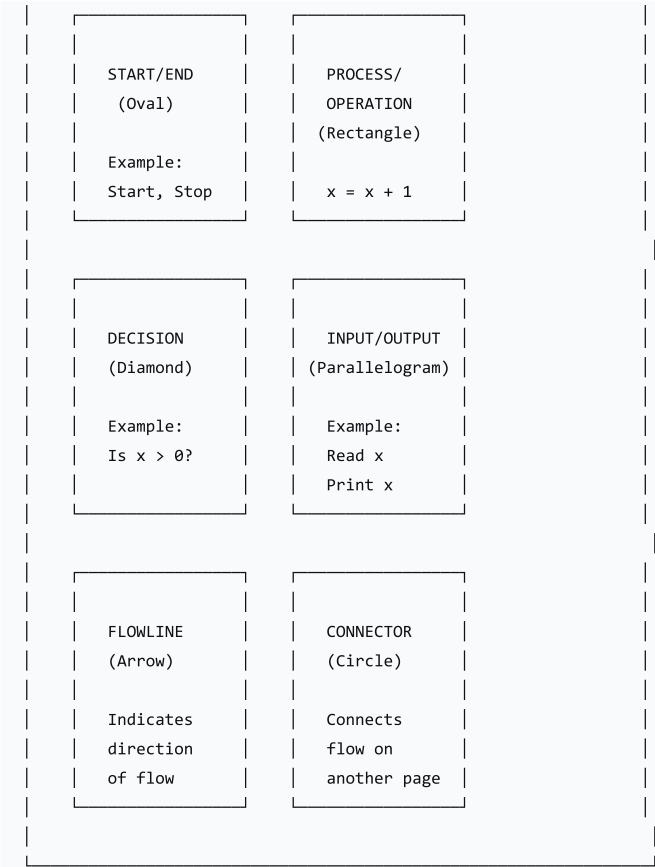
Shares the algorithm with others

3. Standard Flowchart Symbols

The Six Key Symbols

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STANDARD FLOWCHART SYMBOLS



Symbol Summary Table

Symbol	Name	Purpose	Example
Oval	Start/End	Indicates the beginning or end of the algorithm	Start , End
Rectangle	Process/Operation	Performs a calculation or assignment	$x = x + 1$, $sum = sum + num$
Diamond	Decision	Makes a choice (yes/no) based on a condition	Is $x > 0$?
Parallelogram	Input/Output	Reads input or displays output	Read x , Print sum
Arrow	Flowline	Shows the direction of execution	→ , ↓
Circle	Connector	Connects flow on different pages/sections	A , B

4. Flowchart Example 1: Sum of Even Numbers

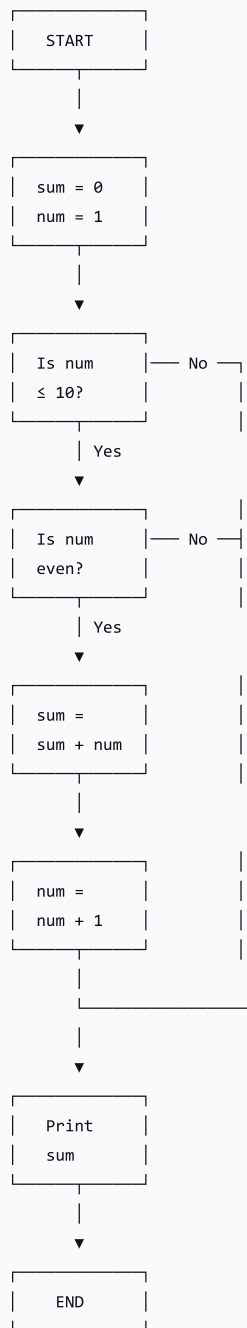
Problem

Calculate the sum of even numbers from 1 to 10.

Flowchart

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FLOWCHART: SUM OF EVEN NUMBERS



5. Trace Tables

What is a Trace Table?

A trace table is a table used to track changes in variable values as a flowchart is executed step by step.

How to Create a Trace Table

Step	Description
1	Identify all variables used in the algorithm
2	Create columns for each variable and an output column
3	Follow the flowchart step by step
4	Record the value of each variable after each step
5	Record the output when the algorithm produces it

Trace Table Example: Sum of Even Numbers

Step	num	sum	Output
1 (Start)	-	-	-
2 (Initialise)	1	0	-
3 (num ≤ 10?)	1	0	-
4 (even?)	1	0	-
5 (num + 1)	2	0	-
6 (num ≤ 10?)	2	0	-
7 (even?)	2	0	-
8 (sum + num)	2	2	-
9 (num + 1)	3	2	-
...	...	...	...
Final	11	30	Output: 30

Completed Trace Table

Step	num	sum	Output
Start	-	-	-
Init	1	0	-
Loop 1	1	0	-
...	...	...	...
End	11	30	30

Output: 30

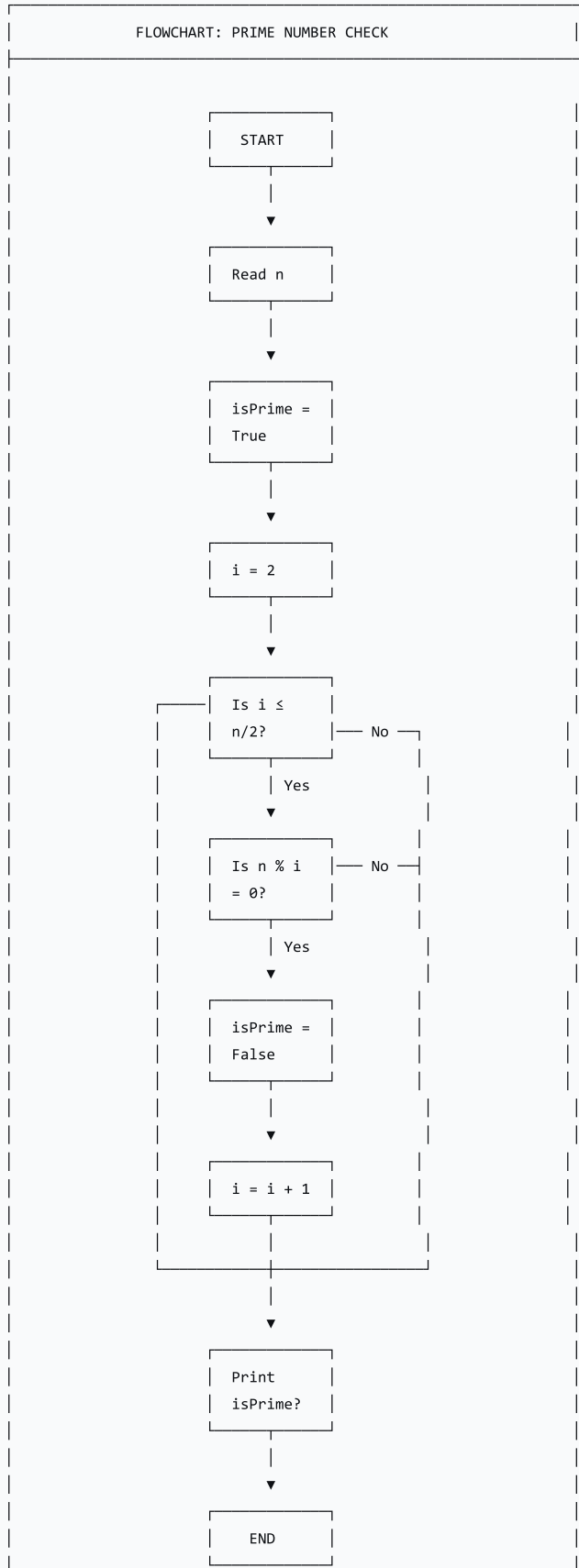
6. Flowchart Example 2: Prime Number Check

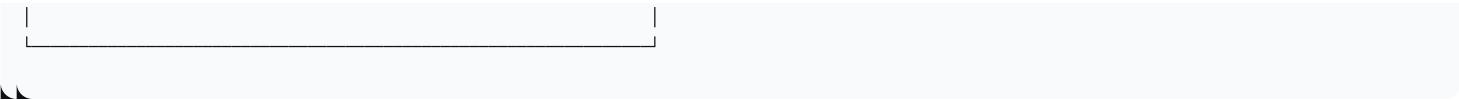
Problem

Determine if a given number is prime.

Flowchart

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7. Activity: Drawing Flowcharts

Instructions

1. Use paper/pencil, Google Drawings, or Canva
2. Create a flowchart for the following algorithms:

Algorithm 1: Sum of Even Numbers from 1 to 10

Steps:

1. Start
2. Initialise sum = 0, num = 1
3. While num ≤ 10:
 - If num is even, add to sum
 - Increment num
4. Print sum
5. End

Algorithm 2: Prime Number Check

Steps:

1. Start
2. Read n
3. Set isPrime = True
4. Set i = 2
5. While i ≤ n/2:
 - If n % i == 0, set isPrime = False
 - Increment i
6. Print isPrime
7. End

8. Tracing Flowcharts: Worksheet Practice

Instructions

1. Trace the execution flow of the given flowchart step by step
2. Keep track of variable values as they change
3. Determine the final output of the algorithm

Practice Flowchart 1

Problem: Calculate the sum of numbers from 1 to 5.

Variable	Initial Value	Changes
sum	0	0, 1, 3, 6, 10, 15
i	1	1, 2, 3, 4, 5, 6

Output: 15

Practice Flowchart 2

Problem: Find the largest number in a list.

List: [3, 7, 2, 9, 4]

Variable	Initial Value	Changes
max	$-\infty$	3, 7, 9

Output: 9

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	Recipe analogy; introduce flowcharts
Standard Symbols	10 min	Explain each symbol with examples
Creating Flowcharts	20 min	Students create flowcharts for given algorithms
Tracing Flowcharts	20 min	Trace execution flow; create trace tables
Discussion	5 min	Challenges and benefits of flowcharts

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Flowchart Creation	Assess ability to draw correct flowcharts
Trace Table Practice	Evaluate ability to trace flowcharts
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

- 1. What symbol is used for a decision in a flowchart?
- 2. What is the purpose of a trace table?
- 3. What is the output of an algorithm that sums numbers 1 to 5?
- 4. Why are flowcharts useful?

11. Differentiation

Level	Strategy
Support	Provide symbol reference sheets; use simple examples first

Level	Strategy
Challenge	Create flowcharts for more complex algorithms (recursion, nested loops)

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (B1.1.4 with examples)
- **Quizlet** (B1.1.4 vocabulary flashcards)
- **Worksheet:** Flowchart tracing

13. Summary: Key Takeaways

Symbol	Name	Purpose
Oval	Start/End	Begins or ends an algorithm
Rectangle	Process	Performs calculations or assignments
Diamond	Decision	Makes a yes/no choice
Parallelogram	Input/Output	Reads or displays data
Arrow	Flowline	Shows direction of flow
Circle	Connector	Connects flow on different pages

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KEY REMINDERS

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Flowcharts visualise algorithms

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Six standard symbols: start, process, decision, input/output, flowline, connector

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Trace tables track variable changes

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Tracing helps determine output

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Flowcharts help design and debug algorithms

"Flowcharts are a powerful visual tool for understanding, designing, and communicating algorithms."